Power Supply Module



File: pwr_module.kicad_sch

CPU Module



File: cpu_module.kicad_sch

WiFi Module



File: wifi_module.kicad_sch

Connector Module



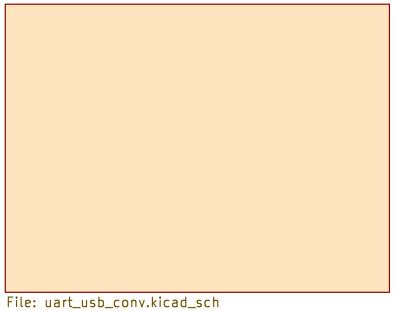
File: conn_module.kicad_sch

Memory Module



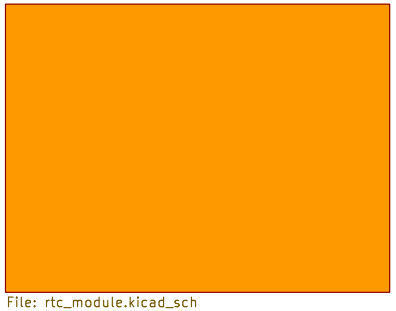
File: mem_module.kicad_sch

UART–USB Converters



File: uart_usb_conv.kicad_sch

RTC Module

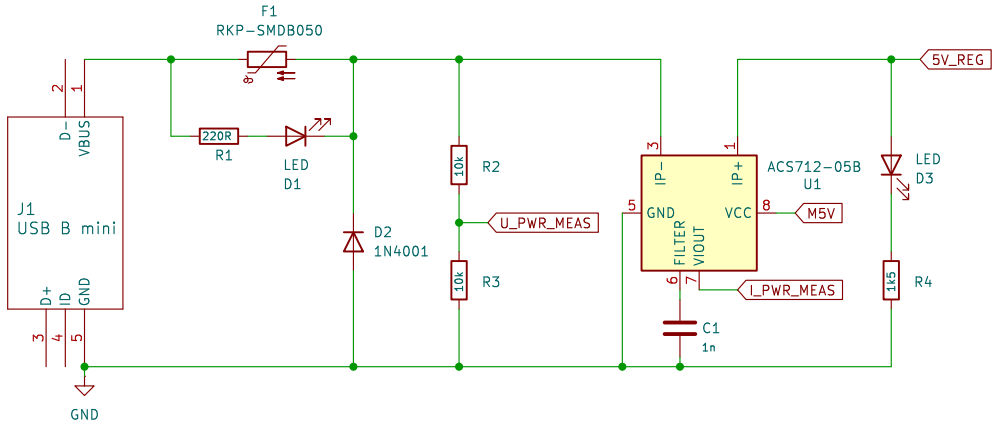


File: rtc_module.kicad_sch

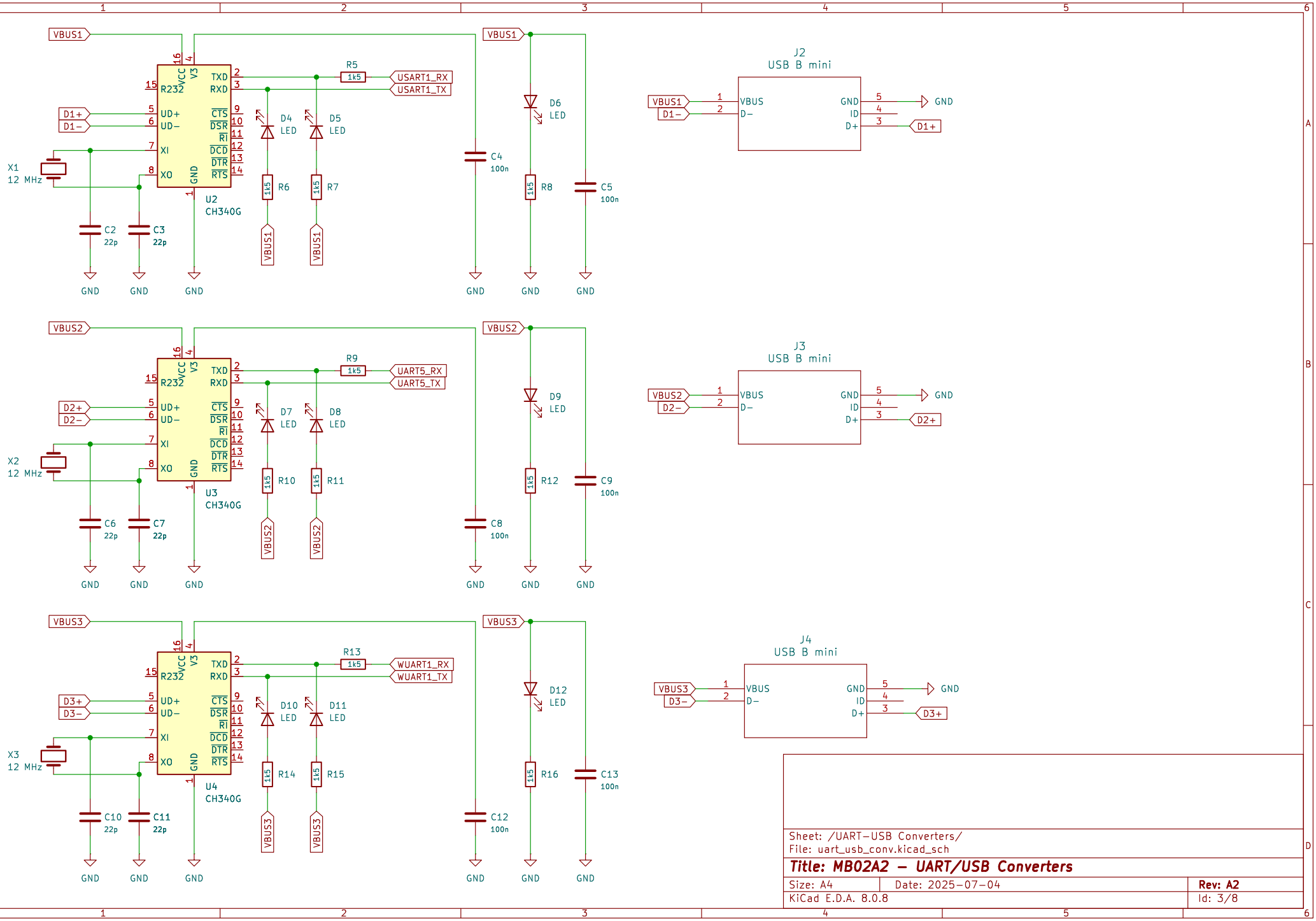
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File: mb02a1.kicad_sch

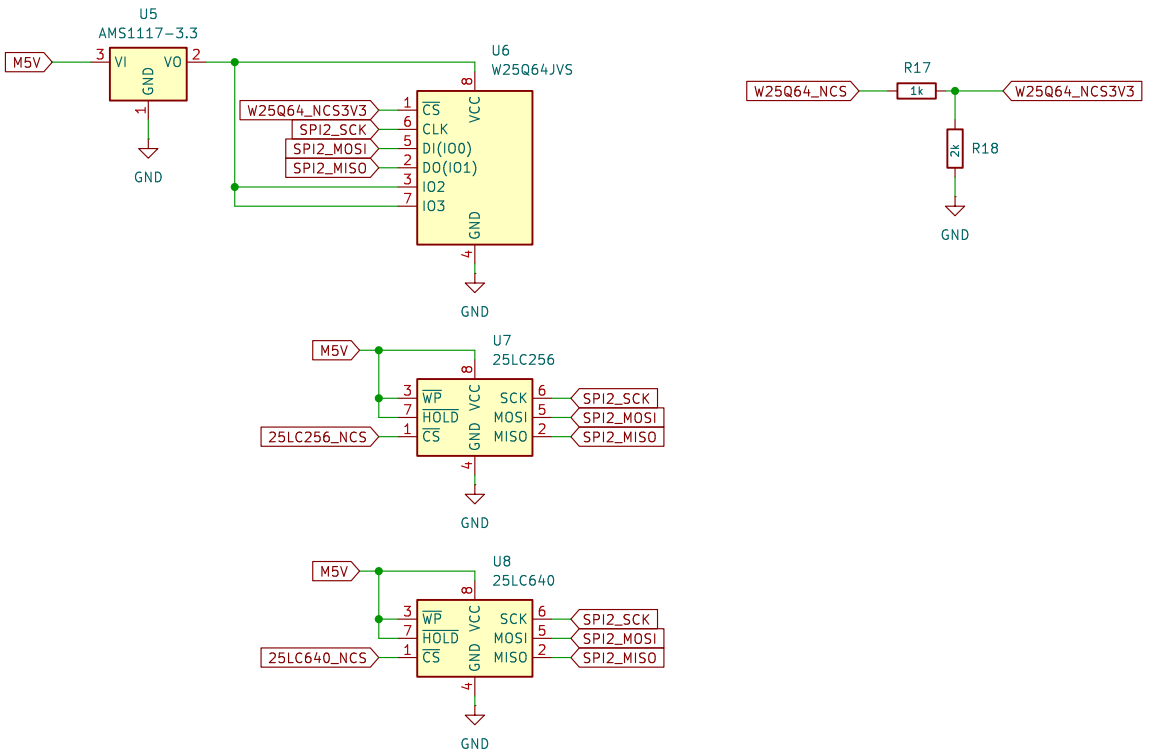
Title: MB02A1

Size: A4	Date: 2025-07-04	**Rev: A2**
KiCad E.D.A. 8.0.8		Id: 1/8

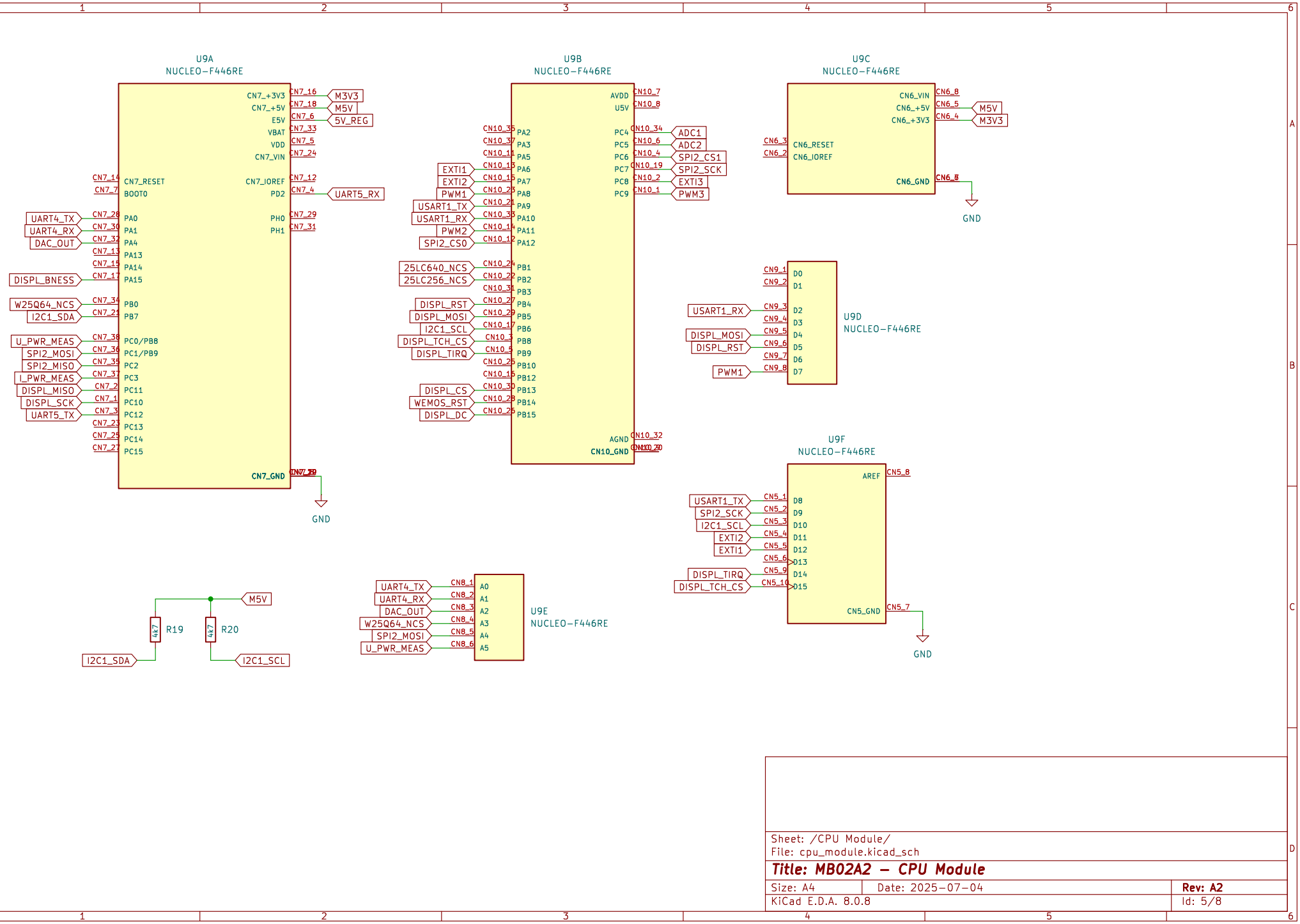


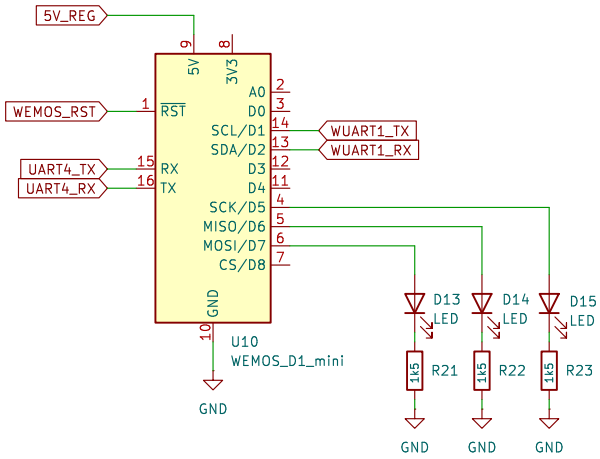
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Title: MB02A2 – Power Supply Module		
Size: A4	Date: 2025-07-04	**Rev: A2**
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VBUS1
U2
CH340G
X1
12 MHz
D1+
D1−
R5
1k5
USART1_RX
USART1_TX
D4
LED
D5
LED
R6
R7
C2
22p
C3
22p
C4
100n
D6
LED
R8
C5
100n
J2
USB B mini
GND
VBUS2
U3
CH340G
X2
12 MHz
D2+
D2−
R9
UART5_RX
UART5_TX
D7
D8
R10
R11
C6
C7
C8
D9
R12
C9
J3
VBUS3
U4
CH340G
X3
12 MHz
D3+
D3−
R13
WUART1_RX
WUART1_TX
D10
D11
R14
R15
C10
C11
C12
D12
R16
C13
J4
Sheet: /UART–USB Converters/
File: uart_usb_conv.kicad_sch
Title: MB02A2 – UART/USB Converters
Size: A4
Date: 2025-07-04
Rev: A2
KiCad E.D.A. 8.0.8
Id: 3/8

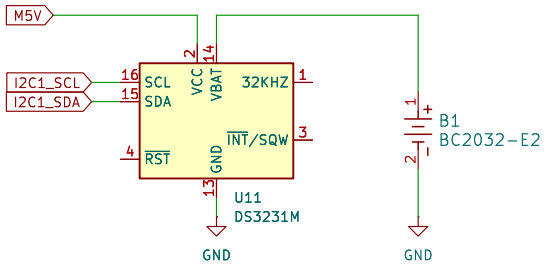


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Size: A4	Date: 2025-07-04	**Rev: A2**
KiCad E.D.A. 8.0.8		Id: 4/8

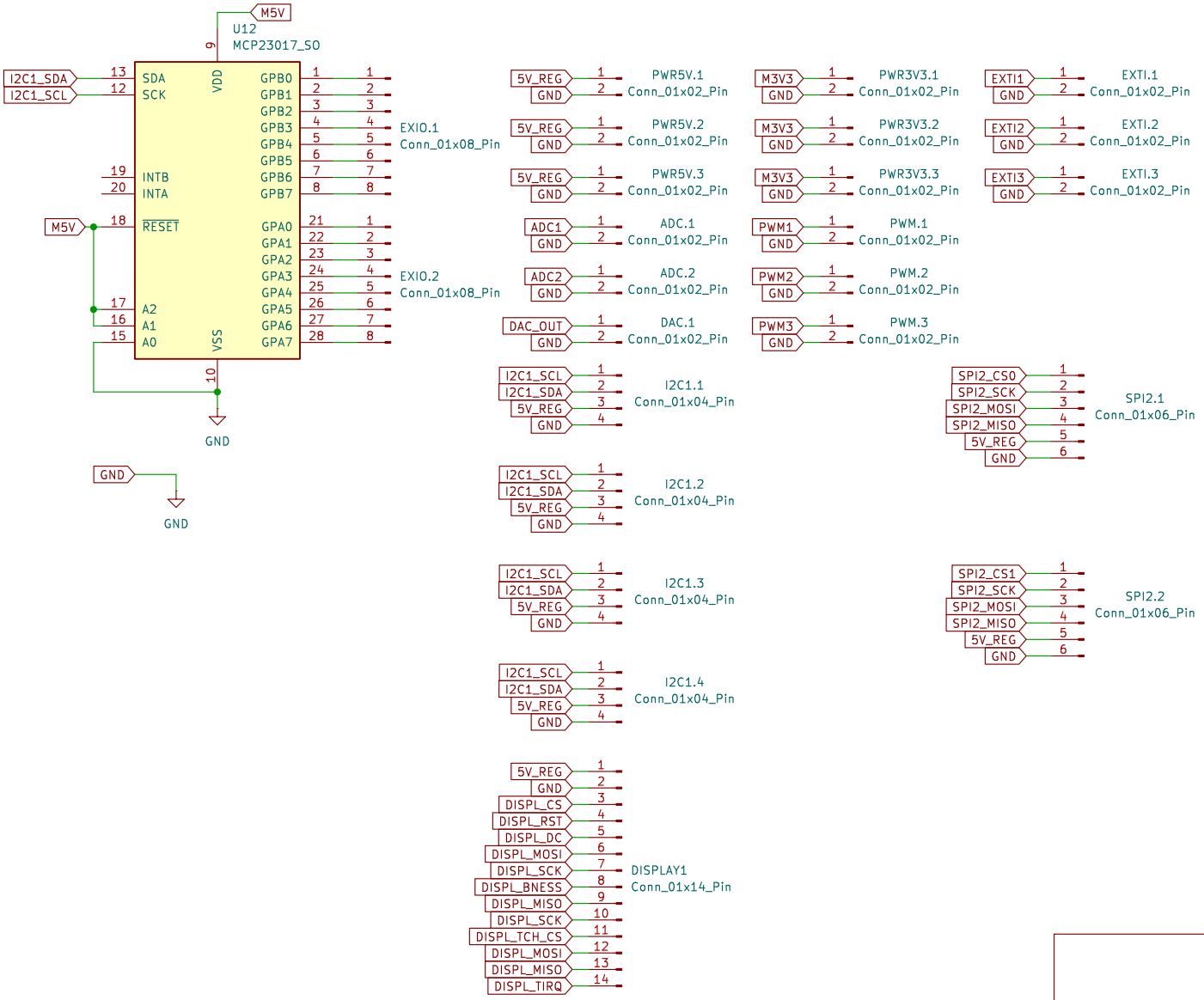

U9A
NUCLEO-F446RE
CN7_+3V3
CN7_+5V
E5V
VBAT
VDD
CN7_VIN
CN7_RESET
BOOT0
CN7_IOREF
PD2
PA0
PA1
PA4
PA13
PA14
PA15
PB0
PB7
PC0/PB8
PC1/PB9
PC2
PC3
PC11
PC10
PC12
PC13
PC14
PC15
PH0
PH1
CN7_GND
M3V3
M5V
5V_REG
UART5_RX
UART4_TX
UART4_RX
DAC_OUT
DISPL_BNESS
W25Q64_NCS
I2C1_SDA
U_PWR_MEAS
SPI2_MOSI
SPI2_MISO
I_PWR_MEAS
DISPL_MISO
DISPL_SCK
UART5_TX
GND
U9B
NUCLEO-F446RE
AVDD
U5V
AGND
CN10_GND
PA2
PA3
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PB1
PB2
PB3
PB4
PB5
PB6
PB8
PB9
PB10
PB12
PB13
PB14
PB15
EXTI1
EXTI2
PWM1
USART1_TX
USART1_RX
PWM2
SPI2_CS0
25LC640_NCS
25LC256_NCS
DISPL_RST
DISPL_MOSI
I2C1_SCL
DISPL_TCH_CS
DISPL_TIRQ
DISPL_CS
WEMOS_RST
DISPL_DC
PC4
PC5
PC6
PC7
PC8
PC9
ADC1
ADC2
SPI2_CS1
SPI2_SCK
EXTI3
PWM3
U9C
NUCLEO-F446RE
CN6_VIN
CN6_+5V
CN6_+3V3
CN6_RESET
CN6_IOREF
CN6_GND
M5V
M3V3
GND
U9D
NUCLEO-F446RE
D0
D1
D2
D3
D4
D5
D6
D7
USART1_RX
DISPL_MOSI
DISPL_RST
PWM1
U9F
NUCLEO-F446RE
AREF
D8
D9
D10
D11
D12
D13
D14
D15
CN5_GND
USART1_TX
SPI2_SCK
I2C1_SCL
EXTI2
EXTI1
DISPL_TIRQ
DISPL_TCH_CS
GND
U9E
NUCLEO-F446RE
A0
A1
A2
A3
A4
A5
UART4_TX
UART4_RX
DAC_OUT
W25Q64_NCS
SPI2_MOSI
U_PWR_MEAS
R19
R20
4k7
M5V
I2C1_SDA
I2C1_SCL
Sheet: /CPU Module/
File: cpu_module.kicad_sch
Title: MB02A2 – CPU Module
Size: A4
Date: 2025-07-04
Rev: A2
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Sheet: /WiFi Module/		
File: wifi_module.kicad_sch		
Title: MB02A2 – WiFi Module		
Size: A4	Date: 2025-07-04	**Rev: A2**
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Sheet: /RTC Module/		
File: rtc_module.kicad_sch		
Title: MB02A2 – RTC Module		
Size: A4	Date: 2025-07-04	**Rev: A2**
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Sheet: /Connector Module/
File: conn_module.kicad_sch

Title: MB02A2 – Connector Module

Size: A4	Date: 2025-07-04	**Rev: A2**
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